

Aurick Anwar

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EDUCATION

McMaster University
Bachelor of Engineering: Engineering Physics

Hamilton, Ontario
Expected 2029

WORK EXPERIENCE

Software Engineering Intern

HermesAI

Toronto, Ontario
May 2026 – August 2026

- Built **React** interfaces for document upload and AI-powered case extraction, supporting legal files exceeding **1,000 pages**.
- Built an **LLM** chatbot feature with **LangChain** that parses, chunks, and extracts case law from legal documents, cutting manual processing time by **50%** for Canadian Law Firms.
- Reduced document processing latency by **30%** by offloading file processing to asynchronous **Node.js**.

Software Engineer

McMaster Drone Team

Hamilton, Ontario
June 2026 – Present

- Built a reproducible **ArduPilot Copter SITL** environment with **Gazebo**, enabling 15+ team members to test autonomous missions without risking flight hardware.
- Automated waypoint missions via **MAVProxy** over **MAVLink**, achieving **[20/20]** successful runs from clean resets with **<1.2 m** waypoint error.
- Injected **GPS-loss failures** to validate failsafes, triggering recovery in **<2 s**, and built **ROS2/pymavlink** tools streaming vehicle telemetry at **20 Hz**.

Software Engineer

McMaster Mars Rover Team

Hamilton, Ontario
September 2026 – Present

- Developed 10+ **ROS2** nodes for perception, localization, and drive control on an **NVIDIA Jetson AGX Orin**, keeping inter-node latency under **3 ms**.
- Implemented ArUco marker detection with **OpenCV**, detecting markers up to **10 m** at **20+ FPS** to guide autonomous waypoint missions.
- Integrated **GPS/IMU** localization into **Nav2**, enabling the rover to navigate to GPS waypoints within **3 m**.

PERSONAL PROJECTS

[NL Robot](#) | Hack the North 2026

[*ROS2, Gazebo, FastAPI, Three.js, ClaudeAPI*]

- Built a robot simulator where users upload a **URDF** and command it in plain English, cutting robot test setup time by **70%** compared with hand-written **ROS2** commands.
- Used the **Claude API** to turn commands into validated plans, executed on things like a **20 Hz ROS loop** controlling a diff-drive base and **4-DOF arm** with **inverse kinematics**.
- Streamed robot state at **30 Hz** through a **FastAPI** bridge to a **Three.js** viewport, and simulated a **360° LiDAR** through raycasting, enabling **GPU-free Gazebo** simulation in **Docker**.

[TeleARM](#)

[*ROS2, SLAM, Gazebo, Nav2, RGB Camera, LiDAR*]

- Built an autonomous **ROS2** pipeline that uses an **RGB camera** to detect different colored boxes from the ground and triggers a forklift-style pickup, reducing manual clean-up by **25%**.
- Integrated a **2D LiDAR** with a **SLAM + Nav2** stack in **Gazebo** for real-time mapping and obstacle avoidance, letting the robot re-route around walls and improving navigation reliability by **50%**.
- Assigned 4 color-coded helipads as destination zones, enabling the robot to autonomously match and deliver each detected box to its corresponding helipad.

Technical Skills

Programming Languages: Python, C++, Javascript, CSS, HTML, MATLAB,

ML and Computer Vision: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv11, LangChain, Mediapipe

Robotics and Simulation: ROS, RVIZ, Gazebo, SLAM, MuJoCo

Electronics and Microcontrollers: Arduino, Raspberry Pi, ESP32

3D Design and CAD: Solidworks, Inventor, Fusion360, AutoCAD

Web Development: Node.js, React

Tools and OS: Linux, Git, Docker